

Suggested problems: 1, 3-9, 13

Problem 1: Suppose that a fair coin is tossed twice. Let A be the event that the second toss is a head, and B the event that the first toss is a head. Compute $P(A | B)$. What do you notice?

Problem 2: (Example 2.2.3) Suppose that a balanced die is rolled. Let A be the event that an even number is obtained, and let B be the event that one of the numbers 1, 2, 3, or 4 is obtained. Show that the events A and B are independent.

Problem 3: (Example 2.2.4) Suppose that a fair coin is tossed twice. Let A be the event that the first toss is a head, B the event that the second toss is a head, and C the event that both tosses are the same. Are A , B , and C independent?

Problem 4: Let D be the event that you are sick with a disease, T the event that a test for the disease shows positive, and V the event that you go on vacation. You are given the following information:

- The probability that you have the disease, the test shows positive, and you go on vacation is 0.144.
- The probability that you have the disease and the test shows positive is 0.72.
- The probability that the test shows positive and you go on vacation is 0.156.
- The probability that the test shows positive is 0.78.

Are the events D that you have the disease and V that you go on vacation conditionally independent given the event T that the test shows positive? Does your answer make sense intuitively?